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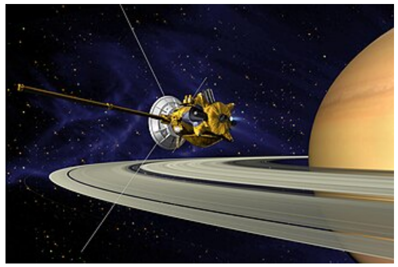
Cassini–Huygens

Cassini–Huygens (/kəˈsiːni ˈhɔɪɡənz/ *kə-SEE-nee HOY-gənz*), commonly called ***Cassini***, was a joint space-research mission by NASA, the European Space Agency (ESA), and the Italian Space Agency (ASI) to send a space probe to study the planet Saturn and its system, including its rings and natural satellites. The Flagship-class robotic spacecraft comprised both NASA's *Cassini* space probe and ESA's *Huygens* lander, which landed on Saturn's largest moon, Titan.[9] *Cassini* was the fourth space probe to visit Saturn and the first to enter its orbit, where it stayed from 2004 to 2017. The two craft took their names from the astronomers Giovanni Cassini and Christiaan Huygens.

Launched aboard a Titan IV-B with a Centaur D-1T upper stage on October 15, 1997, *Cassini* was active in space for nearly 20 years, spending almost 7 years in transit and 13 years orbiting Saturn, studying the planet and its system after entering orbit on July 1, 2004.[10]

The voyage to Saturn included flybys of Venus (April 1998 and July 1999), Earth (August 1999), the asteroid 2685 Masursky, and Jupiter (December 2000). The mission ended on September 15, 2017, when *Cassini*'s trajectory took it into Saturn's upper atmosphere and it burned up[11][12] in order to prevent any risk of contaminating Saturn's moons, which might have offered habitable environments to stowaway terrestrial microbes on the spacecraft.[13][14] The mission was successful beyond expectations – NASA's Planetary Science Division Director, Jim Green, described *Cassini–Huygens* as a "mission of firsts"[15] that revolutionized human understanding of the Saturn system, including its moons and rings, and our understanding of where life might be found in the Solar System.[16]

Cassini–Huygens



Artist's concept of *Cassini*'s orbit insertion around Saturn

Names	Saturn Orbiter and Titan Probe (SOTP)
Mission type	*Cassini*: Saturn orbiter *Huygens*: Titan lander
Operator	*Cassini*: NASA / JPL *Huygens*: ESA / ASI
COSPAR ID	1997-061A
SATCAT no.	25008
Website	▪ NASA (https://saturn.jpl.nasa.gov/) ▪ ESA (https://www.esa.int/Our_Activities/Space_Science/Cassini-Huygens)
Mission duration	**Overall**: 19 years, 335 days 13 years, 76 days at Saturn **En route**: 6 years, 261 days **Prime mission**: 3 years **Extended missions**: *Equinox*: 2 years, 62 days *Solstice*: 6 years, 205 days *Finale*: 4 months, 24 days
Distance travelled	7.9 billion km (4.9 billion mi)[1]
Spacecraft properties	
Manufacturer	*Cassini*: JPL *Huygens*: Aerospatiale[2]
Launch mass	5,712 kg (12,593 lb)[1][3]
Dry mass	2,523 kg (5,562 lb)[1]
Power	~885 watts (at launch)[1] ~670 watts (2010)[4] ~663 watts (at end of mission)[1]
Start of mission	
Launch date	October 15, 1997, 08:43:00 UTC
Rocket	Titan IV-B / Centaur D-1T (B-33)
Launch site	Cape Canaveral, SLC-40
Contractor	Lockheed Martin
End of mission	
Disposal	Controlled entry into Saturn[5][6]
Last contact	September 15, 2017 ▪ 11:55:39 UTC X-band telemetry ▪ 11:55:46 UTC S-band radio science[7]
Orbital parameters	
Reference system	Kronocentric
Flyby of Venus (Gravity assist)	
Closest approach	April 26, 1998
Distance	283 km (176 mi)
Flyby of Venus (Gravity assist)	
Closest approach	June 24, 1999
Distance	623 km (387 mi)
Flyby of Earth-Moon system (Gravity assist)	
Closest approach	August 18, 1999, 03:28 UTC
Distance	1,171 km (728 mi)
Flyby of 2685 Masursky (Incidental)	
Closest approach	January 23, 2000

Overview

Planning and development

Scientists and individuals from 27 countries made up the joint team responsible for designing, building, flying and collecting data from the *Cassini* orbiter and the *Huygens* probe.[16]

Cassini's planners originally scheduled a mission of four years, from June 2004 to May 2008. The mission was extended for another two years until September 2010, branded the *Cassini Equinox Mission*. The mission was extended a second and final time with the *Cassini Solstice Mission*, lasting another seven years until September 15, 2017, on which date *Cassini* was de-orbited to burn up in Saturn's upper atmosphere.[17] The *Huygens* module traveled with *Cassini* until its separation from the probe on December 25, 2004; Huygens landed by parachute.[18] on Titan on January 14, 2005. The separation was facilitated by the SED (Spin/Eject device), which provided a relative separation speed of 0.35 metres per second (1.1 ft/s) and a spin rate of 7.5 rpm.[19] It returned data to Earth for around 90 minutes, using the orbiter as a relay. This was the first landing ever accomplished in the outer Solar System and the first landing on a moon other than Earth's Moon.

At the end of its mission, the *Cassini* spacecraft executed its "Grand Finale": a number of risky passes through the gaps between Saturn and its inner rings.[5][6] This phase aimed to maximize *Cassini*'s scientific outcome before the spacecraft was intentionally destroyed[20] to prevent potential contamination of Saturn's moons if *Cassini* were to unintentionally crash into them when maneuvering the probe was no longer possible due to power loss or other communication issues at the end of its operational lifespan. *Cassini's* atmospheric entry on Saturn ended the mission, but analysis of the returned data will continue for many years.[16]

Distance	1,600,000 km (990,000 mi)
Flyby of Jupiter (Gravity assist)	
Closest approach	December 30, 2000
Distance	9,852,924 km (6,122,323 mi)
Saturn orbiter	
Spacecraft component	*Cassini*
Orbital insertion	July 1, 2004, 02:48 UTC
Orbits	294[1]
Titan lander	
Spacecraft component	*Huygens*
Landing date	January 14, 2005
Landing site	10.573°S 192.335°W[8]
Large Strategic Science Missions	
Planetary Science Division	

NASA and JPL

NASA's Jet Propulsion Laboratory, which developed the orbiter, managed the mission. The European Space Research and Technology Centre developed *Huygens*. The centre's prime contractor, Aérospatiale of France (which became part of Thales Alenia Space in 2005), assembled the probe with equipment and instruments supplied by many European countries (including *Huygens*'s batteries and two scientific instruments from the United States). The Italian Space Agency (ASI) provided the *Cassini* orbiter's high-gain radio antenna, with the incorporation of a low-gain antenna (to ensure telecommunications with the Earth for the entire duration of the mission), a compact and lightweight radar, which also used the high-gain antenna and served as a synthetic-aperture radar, a radar altimeter, a radiometer, the radio science subsystem (RSS), and the visible-channel portion VIMS-V of VIMS spectrometer.[21]

The 90s were difficult years for NASA and JPL: with the Cold War and the Space Race between the US and the USSR finished, NASA saw dwindling budgets and introduced the so-called "faster, better, cheaper" approach, that encouraged smaller, cheaper missions built with the help of third-party contractors, efficiently "commercializing" the research lab and forcing it to work with industry.[22]:208,254 Caltech leadership even feared that JPL could be closed.[22]:208–209

JPL had little experience in small missions at the time: its "flagship" missions, like *Voyager*, *Cassini*, and *Galileo*, employed hundreds of people for decades. Cassini "directly supported maybe 500 work-years, about 10 percent of total lab staff [and] provided close to 20 percent of the lab budget".[22]:223 In order to comply with budget restrictions and save the mission, *Cassini* was downsized; to save $250 million, the scan platform had to be removed from the plans.[22]:260–261

NASA provided the VIMS infrared counterpart, as well as the Main Electronic Assembly, which included electronic sub-assemblies provided by CNES of France.[23][24] On April 16, 2008, NASA announced a two-year extension of the funding for ground operations of this mission, at which point it was renamed the *Cassini* Equinox Mission.[25] It was extended again in February 2010 as the *Cassini* Solstice Mission.

Naming

The mission consisted of two main elements: the ASI/NASA *Cassini* orbiter, named for the Italian astronomer Giovanni Domenico Cassini, the discoverer of Saturn's ring divisions and four of its satellites; and the ESA-developed *Huygens* probe, named for the Dutch astronomer, mathematician and physicist Christiaan Huygens, discoverer of Titan.

The mission was commonly called Saturn Orbiter Titan Probe (SOTP) during gestation, both as a Mariner Mark II mission and generically.[26]

Cassini–Huygens was a *Flagship*-class mission to the outer planets.[9] The other planetary flagships include *Galileo*, Voyager, and Viking.[9]

Objectives

Cassini had several objectives, including:[27]

- Determining the three-dimensional structure and dynamic behavior of the rings of Saturn.
- Determining the composition of the satellite surfaces and the geological history of each object.
- Determining the nature and origin of the dark material on Iapetus's leading hemisphere.
- Measuring the three-dimensional structure and dynamic behavior of the magnetosphere.
- Studying the dynamic behavior of Saturn's atmosphere at cloud level.
- Studying the time variability of Titan's clouds and hazes.
- Characterizing Titan's surface on a regional scale.

Cassini–Huygens was launched on October 15, 1997, from Cape Canaveral Air Force Station's Space Launch Complex 40 using a U.S. Air Force Titan IVB/Centaur rocket. The complete launcher was made up of a two-stage Titan IV booster rocket, two strap-on solid rocket engines, the Centaur upper stage, and a payload enclosure, or fairing.[28]

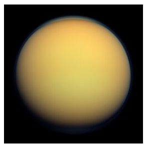
The total cost of this scientific exploration mission was about US$3.26 billion, including $1.4 billion for pre-launch development, $704 million for mission operations, $54 million for tracking and $422 million for the launch vehicle. The United States contributed $2.6 billion (80%), the ESA $500 million (15%), and the ASI $160 million (5%).[29] However, these figures are from the press kit which was prepared in October 2000. They do not include inflation over the course of a very long mission, nor do they include the cost of the extended missions.

The primary mission for *Cassini* was completed on July 30, 2008. The mission was extended to June 2010 (*Cassini* Equinox Mission).[30] This studied the Saturn system in detail during the planet's equinox, which happened in August 2009.[25]

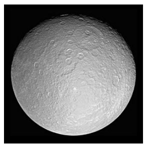
On February 3, 2010, NASA announced another extension for *Cassini*, lasting 6½ years until 2017, ending at the time of summer solstice in Saturn's northern hemisphere (*Cassini* Solstice Mission). The extension enabled another 155 revolutions around the planet, 54 flybys of Titan and 11 flybys of Enceladus.[31] In 2017, an encounter with Titan changed its orbit in such a way that, at closest approach to Saturn, it was only 3,000 km (1,900 mi) above the planet's cloudtops, below the inner edge of the D ring. This sequence of "proximal orbits" ended when its final encounter with Titan sent the probe into Saturn's atmosphere to be destroyed.

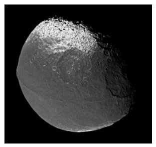
Destinations

Selected destinations (ordered largest to smallest but not to scale)

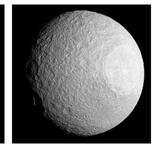

Titan

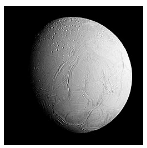

Earth's Moon

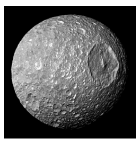

Rhea


Iapetus


Dione


Tethys

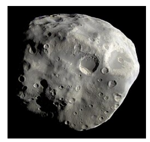

Enceladus

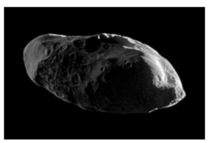

Mimas

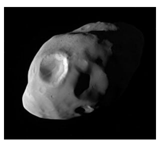

Hyperion

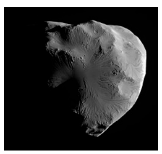

Phoebe

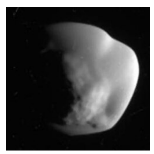

Janus

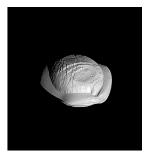

Epimetheus

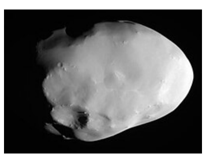

Prometheus

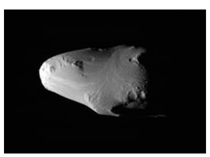

Pandora

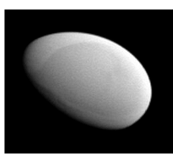

Helene


Atlas


Pan

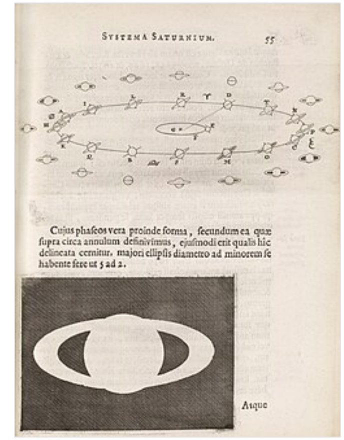

Telesto


Calypso


Methone

History

Cassini–Huygens's origins date to 1982, when the European Science Foundation and the American National Academy of Sciences formed a working group to investigate future cooperative missions. Two European scientists suggested a paired Saturn Orbiter and Titan Probe as a possible joint mission. In 1983, NASA's *Solar System Exploration Committee* recommended the same Orbiter and Probe pair as a core NASA project. NASA and the European Space Agency (ESA) performed a joint study of the potential mission from 1984 to 1985. ESA continued with its own study in 1986, while the American astronaut Sally Ride, in her influential 1987 report *NASA Leadership and America's Future in Space*, also examined and approved of the *Cassini* mission.[32]


SYSTEMA SATURNIUM. 55

Cujus phaseos vera proinde forma, secundum ea quæ supra circa annulum definivimus, ejusmodi erit qualis hic delineata cernitur, majori ellipsis diametro ad minorem se habente fere ut 5 ad 2.

Atque

Huygens's explanation for the aspects of Saturn, *Systema Saturnium* (1659)

While Ride's report described the Saturn orbiter and probe as a NASA solo mission, in 1988 the Associate Administrator for Space Science and Applications of NASA, Len Fisk, returned to the idea of a joint NASA and ESA mission. He wrote to his counterpart at ESA, Roger Bonnet, strongly suggesting that ESA choose the *Cassini* mission from the three candidates at hand and promising that NASA would commit to the mission as soon as ESA did.[33]

At the time, NASA was becoming more sensitive to the strain that had developed between the American and European space programs as a result of European perceptions that NASA had not treated it like an equal during previous collaborations. NASA officials and advisers involved in promoting and planning *Cassini–Huygens* attempted to correct this trend by stressing their desire to evenly share any scientific and technology benefits resulting from the mission. In part, this newfound spirit of cooperation with Europe was driven by a sense of competition with the Soviet Union, which had begun to cooperate more closely with Europe as ESA drew further away from NASA. Late in 1988, ESA chose Cassini–Huygens as its next major mission and the following year the program received major funding in the US.[34][35]


A Titan IV-B rocket and Centaur-T upper stage with the *Cassini–Huygens* payload at LC-40, three days before its 15 October 1997 launch

The collaboration not only improved relations between the two space programs but also helped *Cassini–Huygens* survive congressional budget cuts in the United States. *Cassini–Huygens* came under fire politically in both 1992 and 1994, but NASA successfully persuaded the United States Congress that it would be unwise to halt the project after ESA had already poured funds into development because frustration on broken space exploration promises might spill over into other areas of foreign relations. The project proceeded politically smoothly after 1994, although citizens' groups concerned about the potential environmental impact a launch failure might have (because of its plutonium power source) attempted to derail it through protests and lawsuits until and past its 1997 launch.[36][37][38][39][40]

Spacecraft design

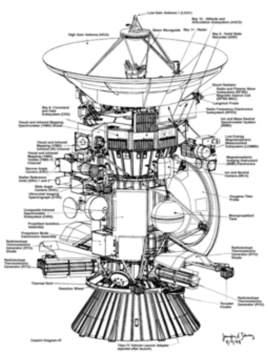


Diagram of Cassini


Cassini–Huygens assembly

The spacecraft was planned to be the second three-axis stabilized, RTG-powered Mariner Mark II, a class of spacecraft developed for missions beyond the orbit of Mars, after the Comet Rendezvous Asteroid Flyby (CRAF) mission, but budget cuts and project rescopings forced NASA to terminate CRAF development to save *Cassini*. As a result, *Cassini* became more specialized. The Mariner Mark II series was cancelled.

The combined orbiter and probe was at the time the third-largest uncrewed interplanetary spacecraft ever successfully launched, behind the Phobos 1 and 2 Mars probes, as well as being among the most complex;[41][42] NASA's *Europa Clipper* became the new third-largest probe upon its launch in 2024.[43] The orbiter had a mass of 2,150 kg (4,740 lb), the probe 350 kg (770 lb) including 30 kg (66 lb) of probe support equipment left on the orbiter. With the launch vehicle adapter and 3,132 kg (6,905 lb) of propellants at launch, the spacecraft had a mass of 5,600 kg (12,300 lb).


Cassini–Huygens original design (Mariner Mark II, 1988)

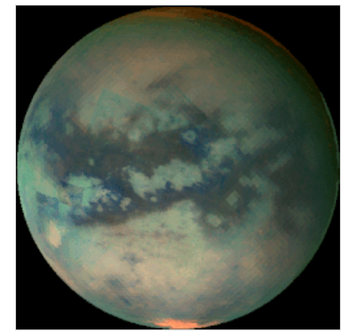
The *Cassini* spacecraft was 6.8 meters (22 ft) high and 4 meters (13 ft) wide. Its bus was a dodecagonal prism atop a conical frustum connecting it to a cylinder containing the propellant tanks, to which the RTGs and *Huygens* were attached.[44] Spacecraft complexity was increased by its trajectory (flight path) to Saturn, and by the ambitious science at its destination. *Cassini* had 1,630 interconnected electronic components, 22,000 wire connections, and 14 kilometers (8.7 mi) of cabling.[45] The core control computer CPU was a redundant system using the MIL-STD-1750A instruction set architecture. The main propulsion system consisted of one prime and one backup R-4D bipropellant rocket engine. The thrust of each engine was 490 N (110 lbf) and the total spacecraft delta-v was 2,352 m/s (5,260 mph).[46] Smaller monopropellant rockets provided attitude control.

Cassini was powered by 32.7 kg (72 lb) of nuclear fuel, mainly plutonium dioxide (containing 28.3 kg (62 lb) of pure plutonium).[47] The heat from the material's radioactive decay was turned into electricity. *Huygens* was supported by *Cassini* during cruise, but used chemical batteries when independent.

The probe contained a DVD with more than 616,400 signatures from citizens in 81 countries, collected in a public campaign.[48][49]

Until September 2017 the *Cassini* probe continued orbiting Saturn at a distance of between 8.2 and 10.2 astronomical units (1.23×10^9 and 1.53×10^9 km; 760,000,000 and 950,000,000 mi) from the Earth. It took 68 to 84 minutes for radio signals to travel from Earth to the spacecraft, and vice versa. Thus ground controllers could not give "real-time" instructions for daily operations or for unexpected events. Even if response were immediate, more than two hours would have passed between the occurrence of a problem and the reception of the engineers' response by the satellite.

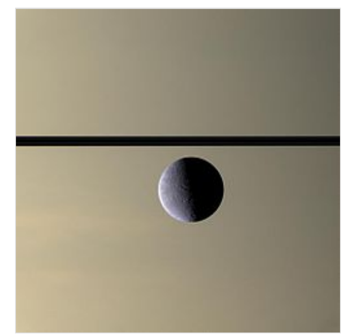
Instruments


Titan's surface revealed by VIMS

Summary

Instruments:[51]

- Optical Remote Sensing ("Located on the remote sensing pallet")[51]
  - Composite Infrared Spectrometer (CIRS)
  - Imaging Science Subsystem (ISS)
  - Ultraviolet Imaging Spectrograph (UVIS)
  - Visual and Infrared Mapping Spectrometer (VIMS)
- Fields, Particles and Waves (mostly in situ)
  - Cassini Plasma Spectrometer (CAPS)
  - Cosmic Dust Analyzer (CDA)
  - Ion and Neutral Mass Spectrometer (INMS)
  - Magnetometer (MAG)
  - Magnetospheric Imaging Instrument (MIMI)
  - Radio and Plasma Wave Science (RPWS)
- Microwave Remote Sensing
  - Radar
  - Radio Science (RSS)

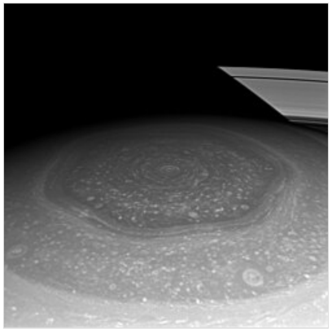

Rhea in front of Saturn

Description

Cassini's instrumentation consisted of: a synthetic aperture radar mapper, a charge-coupled device imaging system, a visible/infrared mapping spectrometer, a composite infrared spectrometer, a cosmic dust analyzer, a radio and plasma wave experiment, a plasma spectrometer, an ultraviolet imaging spectrograph, a magnetospheric imaging instrument, a magnetometer and an ion/neutral mass spectrometer. Telemetry from the communications antenna and other special transmitters (an S-band transmitter and a dual-frequency K_a-band system) was also used to make observations of the atmospheres of Titan and Saturn and to measure the gravity fields of the planet and its satellites.

Cassini Plasma Spectrometer (CAPS)

CAPS was an in situ instrument that measured the flux of charged particles at the location of the spacecraft, as a function of direction and energy. The ion composition was also measured using a time-of-flight mass spectrometer. CAPS measured particles produced by ionisation of molecules originating from Saturn's and Titan's ionosphere, as well as the plumes of Enceladus. CAPS also investigated plasma in these areas, along with the solar wind and its interaction with Saturn's magnetosphere.[51][52] CAPS was turned off in June 2011, as a precaution due to a "soft" electrical short circuit that occurred in the instrument. It was powered on again in March 2012, but after 78 days another short circuit forced the instrument to be shut down permanently.[53]



Saturn's north polar hexagon[50]

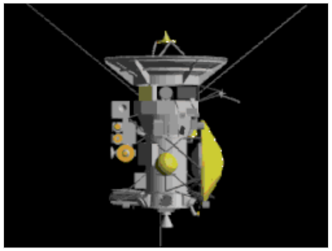
Cosmic Dust Analyzer (CDA)

The CDA was an in situ instrument that measured the size, speed, and direction of tiny dust grains near Saturn. It could also measure the grains' chemical elements.[54] Some of these particles orbited Saturn, while others came from other star systems. The CDA on the orbiter was designed to learn more about these particles, the materials in other celestial bodies and potentially about the origins of the universe.[51]

Saturn in natural-color (January 2010)

Composite Infrared Spectrometer (CIRS)

The CIRS was a remote sensing instrument that measured the infrared radiation coming from objects to learn about their temperatures, thermal properties, and compositions. Throughout the *Cassini–Huygens* mission, the CIRS measured infrared emissions from atmospheres, rings and surfaces in the vast Saturn system. It mapped the atmosphere of Saturn in three dimensions to determine temperature and pressure profiles with altitude, gas composition, and the distribution of aerosols and clouds. It also measured thermal characteristics and the composition of satellite surfaces and rings.[51]



Animated 3D model of the spacecraft

Ion and Neutral Mass Spectrometer (INMS)

The INMS was an in situ instrument that measured the composition of charged particles (protons and heavier ions) and neutral particles (atoms and molecules) near Titan and Saturn to learn more about their atmospheres. The instrument used a quadrupole mass spectrometer. INMS was also intended to measure the positive ion and neutral environments of Saturn's icy satellites and rings.[51][55][56]

Imaging Science Subsystem (ISS)

The ISS was a remote sensing instrument that captured most images in visible light, and also some infrared images and ultraviolet images. The ISS took hundreds of thousands of images of Saturn, its rings, and its moons. The ISS had both a wide-angle camera (WAC) and a narrow-angle camera (NAC). Each of these cameras used a sensitive charge-coupled device (CCD) as its electromagnetic wave detector. Each CCD had a 1,024x1,024 square array of pixels, each pixel 12 μm square. Both cameras allowed for many data collection modes, including on-chip data compression, and were fitted with spectral filters that rotated on a wheel to view different bands within the electromagnetic spectrum ranging from 0.2 to 1.1 μm.[51][57]

Dual Technique Magnetometer (MAG)

The MAG was an in situ instrument that measured the strength and direction of the magnetic field around Saturn. The magnetic fields are generated partly by the molten core at Saturn's center. Measuring the magnetic field is one of the ways to probe the core. MAG aimed to develop a three-dimensional model of Saturn's magnetosphere, and determine the magnetic state of Titan and its atmosphere, and the icy satellites and their role in the magnetosphere of Saturn.[51][58]

Magnetospheric Imaging Instrument (MIMI)

The MIMI was both an in situ and remote sensing instrument that produces images and other data about the particles trapped in Saturn's huge magnetic field, or magnetosphere. The in situ component measured energetic ions and electrons while the remote sensing component (the Ion And Neutral Camera, INCA) was an energetic neutral atom imager.[59] This information was used to study the overall configuration and dynamics of the magnetosphere and its interactions with the solar wind, Saturn's atmosphere, Titan, rings, and icy satellites.[51][60]

Radar

The on-board radar was an active and passive sensing instrument that produced maps of Titan's surface. Radar waves were powerful enough to penetrate the thick veil of haze surrounding Titan. By measuring the send and return time of the signals it is possible to determine the height of large surface features, such as mountains and canyons. The passive radar listened for radio waves that Saturn or its moons may emit.[51]

Radio and Plasma Wave Science instrument (RPWS)

The RPWS was an in situ instrument and remote sensing instrument that receives and measures radio signals coming from Saturn, including the radio waves given off by the interaction of the solar wind with Saturn and Titan. RPWS measured the electric and magnetic wave fields in the interplanetary medium and planetary magnetospheres. It also determined the electron density and temperature near Titan and in some regions of Saturn's magnetosphere using either plasma waves at characteristic frequencies (e.g. the upper hybrid line) or a Langmuir probe. RPWS studied the configuration of Saturn's magnetic field and its relationship to Saturn Kilometric Radiation (SKR), as well as monitoring and mapping Saturn's ionosphere, plasma, and lightning from Saturn's (and possibly Titan's) atmosphere.[51]

Radio Science Subsystem (RSS)

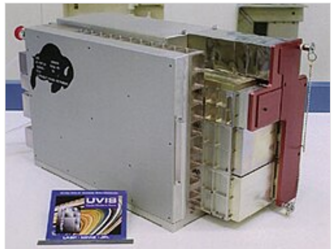
The RSS was a remote-sensing instrument that used radio antennas on Earth to observe the way radio signals from the spacecraft changed as they were sent through objects, such as Titan's atmosphere or Saturn's rings, or even behind the Sun. The RSS also studied the compositions, pressures and temperatures of atmospheres and ionospheres, radial structure and particle size distribution within rings, body and system masses and the gravitational field. The instrument used the spacecraft X-band communication link as well as S-band downlink and K_a-band uplink and downlink.[51]

Ultraviolet Imaging Spectrograph (UVIS)

The UVIS was a remote-sensing instrument that captured images of the ultraviolet light reflected off an object, such as the clouds of Saturn and/or its rings, to learn more about their structure and composition. Designed to measure ultraviolet light over wavelengths from 55.8 to 190 nm, this instrument was also a tool to help determine the composition, distribution, aerosol particle content and temperatures of their atmospheres. Unlike other types of spectrometer, this sensitive instrument could take both spectral and spatial readings. It was particularly adept at determining the composition of gases. Spatial observations took a wide-by-narrow view, only one pixel tall and 64 pixels across. The spectral dimension was 1,024 pixels per spatial pixel. It could also take many images that create movies of the ways in which this material is moved around by other forces.[51]

UVIS consisted of four separate detector channels, the Far Ultraviolet (FUV), Extreme Ultraviolet (EUV), High Speed Photometer (HSP) and the Hydrogen-Deuterium Absorption Cell (HDAC). UVIS collected hyperspectral imagery and discrete spectra of Saturn, its moons and its rings, as well as stellar occultation data.[61]

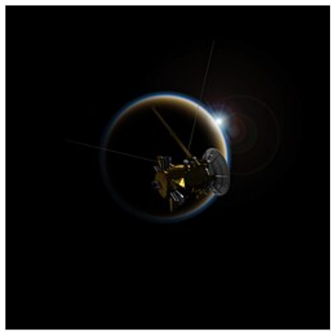
The HSP channel is designed to observe starlight that passes through Saturn's rings (known as stellar occultations) in order to understand the structure and optical depth of the rings.[62] Stellar occultation data from both the HSP and FUV channels confirmed the existence of water vapor plumes at the south pole of Enceladus, as well as characterized the composition of the plumes.[63]



Cassini UVIS instrument built by the Laboratory for Atmospheric and Space Physics at the University of Colorado.

Visible and Infrared Mapping Spectrometer (VIMS)

The VIMS was a remote sensing instrument that captured images using visible and infrared light to learn more about the composition of moon surfaces, the rings, and the atmospheres of Saturn and Titan. It consisted of two cameras - one used to measure visible light, the other infrared. VIMS measured reflected and emitted radiation from atmospheres, rings and surfaces over wavelengths from 350 to 5100 nm, to help determine their compositions, temperatures and structures. It also observed the sunlight and starlight that passes through the rings to learn more about their structure. Scientists used VIMS for long-term studies of cloud movement and morphology in the Saturn system, to determine Saturn's weather patterns.[51]

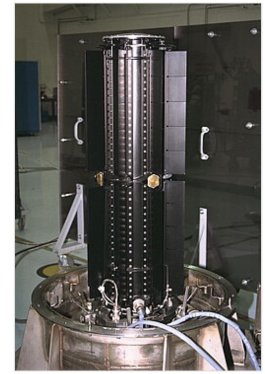


VIMS spectra taken while looking through Titan's atmosphere towards the Sun helped understand the atmospheres of exoplanets (artist's concept; May 27, 2014).

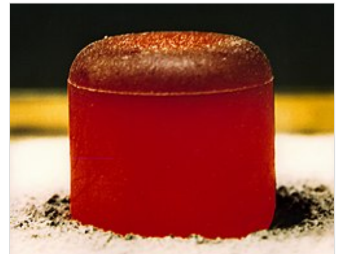
Plutonium power source

Because of Saturn's distance from the Sun, solar arrays were not feasible as power sources for this space probe.[64] To generate enough power, such arrays would have been too large and too heavy.[64] Instead, the *Cassini* orbiter was powered by three GPHS-RTG radioisotope thermoelectric generators, which use heat from the decay of about 33 kg (73 lb) of plutonium-238 (in the form of plutonium dioxide) to generate direct current electricity via thermoelectrics.[64] The RTGs on the *Cassini* mission have the same design as those used on the *New Horizons*, *Galileo*, and *Ulysses* space probes, and they were designed to have very long operational lifetimes.[64] At the end of the nominal 11-year *Cassini* mission, they were still able to produce 600 to 700 watts of electrical power.[64] (Leftover hardware from the *Cassini* RTG Program was modified and used to power the *New Horizons* mission to Pluto and the Kuiper belt, which was designed and launched later.[65])

Electric power distribution was accomplished by 192 solid-state power switches, which also functioned as circuit breakers in the event of an overload condition. The switches used MOSFETs that featured better efficiency and a longer lifetime as compared to conventional switches, while at the same time eliminating transients. However, these solid-state circuit breakers were prone to erroneous tripping (presumably from cosmic rays), requiring them to reset and causing losses in experimental data.[66]



A *Cassini* GPHS-RTG before installation



A glowing-hot plutonium pellet was the power source for the probe's radioisotope thermoelectric generator

To gain momentum while already in flight, the trajectory of the *Cassini* mission included several gravitational slingshot maneuvers: two fly-by passes of Venus, one more of the Earth, and then one of the planet Jupiter. The terrestrial flyby was the final instance when the probe posed any conceivable danger to human beings. The maneuver was successful, with *Cassini* passing by 1,171 km (728 mi) above the Earth on August 18, 1999.[1] Had there been any malfunction causing the probe to collide with the Earth, NASA's complete environmental impact study estimated that, in the worst case (with an acute angle of entry in which *Cassini* would gradually burn up), a significant fraction of the 33 kg[47] of nuclear fuel inside the RTGs would have been dispersed into the Earth's atmosphere so that up to five billion people (i.e. almost the entire terrestrial population) could have been exposed, causing up to an estimated 5,000 additional cancer deaths over the subsequent decades[67] (0.0005 per cent, i.e. a fraction 0.000005, of a billion cancer deaths expected anyway from other causes; the product is incorrectly calculated elsewhere[68] as 500,000 deaths). However, the chance of this happening were estimated to be less than one in one million, i.e. a chance of one person dying (assuming 5,000 deaths) as less than 1 in 200.[67]

NASA's risk analysis to use plutonium was publicly criticized by Michio Kaku on the grounds that casualties, property damage, and lawsuits resulting from a possible accident, as well as the potential use of other energy sources, such as solar and fuel cells, were underestimated.[69]

Telemetry

The *Cassini* spacecraft was capable of transmitting in several different telemetry formats. The telemetry subsystem is perhaps the most important subsystem, because without it there could be no data return.

The telemetry was developed from the ground up, due to the spacecraft using a more modern set of computers than previous missions.[70] Therefore, *Cassini* was the first spacecraft to adopt mini-packets to reduce the complexity of the Telemetry Dictionary, and the software development process led to the creation of a Telemetry Manager for the mission.

There were around 1088 channels (in 67 mini-packets) assembled in the *Cassini* Telemetry Dictionary. Out of these 67 lower complexity mini-packets, 6 mini-packets contained the subsystem covariance and Kalman gain elements (161 measurements), not used during normal mission operations. This left 947 measurements in 61 mini-packets.

A total of seven telemetry maps corresponding to 7 AACS telemetry modes were constructed. These modes are: (1) Record; (2) Nominal Cruise; (3) Medium Slow Cruise; (4) Slow Cruise; (5) Orbital Ops; (6) Av; (7) ATE (Attitude Estimator) Calibration. These 7 maps cover all spacecraft telemetry modes.

Huygens probe

Huygens view of Titan's surface

Same image with different data processing

The *Huygens* probe, supplied by the European Space Agency (ESA) and named after the 17th century Dutch astronomer who first discovered Titan, Christiaan Huygens, scrutinized the clouds, atmosphere, and surface of Saturn's moon Titan in its descent on January 15, 2005. It was designed to enter and brake in Titan's atmosphere and parachute a fully instrumented robotic laboratory down to the surface.[71]

The probe system consisted of the probe itself which descended to Titan, and the probe support equipment (PSE) which remained attached to the orbiting spacecraft. The PSE includes electronics that track the probe, recover the data gathered during its descent, and process and deliver the data to the orbiter that transmits it to Earth. The core control computer CPU was a redundant MIL-STD-1750A control system.

The data was transmitted by a radio link between *Huygens* and *Cassini* provided by Probe Data Relay Subsystem (PDRS). As the probe's mission could not be telecommanded from Earth because of the great distance, it was automatically managed by the Command Data Management Subsystem (CDMS). The PDRS and CDMS were provided by the Italian Space Agency (ASI).

After *Cassini*'s launch, it was discovered that data sent from the *Huygens* probe to *Cassini* orbiter (and then re-transmitted to Earth) would be largely unreadable. The cause was that the bandwidth of signal processing electronics was too narrow and the anticipated Doppler shift between the lander and the mother craft would put the signals out of the system's range. Thus, *Cassini*'s receiver would be unable to receive the data from *Huygens* during its descent to Titan.[20]

A work-around was found to recover the mission. The trajectory of *Cassini* was altered to reduce the line of sight velocity and therefore the doppler shift.[20][72] *Cassini*'s subsequent trajectory was identical to the previously planned one, although the change replaced two orbits prior to the *Huygens* mission with three, shorter orbits.